



While managing date palm plantations in Egypt, Prospero Alpini observed the difference between the sexes of plants.

“THE MARVELLOUS PROPERTY OF THE PENDULUM IS THAT IT MAKES ALL ITS VIBRATIONS... IN EQUAL TIMES.”

Galileo Galilei, Italian astronomer and physicist

English explorer and navigator **Steven Borough** (1525–84) had previously organized the English translation of the standard textbook on navigation of the time: *Breve Compendio* (Brief Summary) or *Arte de Navegar* (Art of Sailing) by Martín Cortés

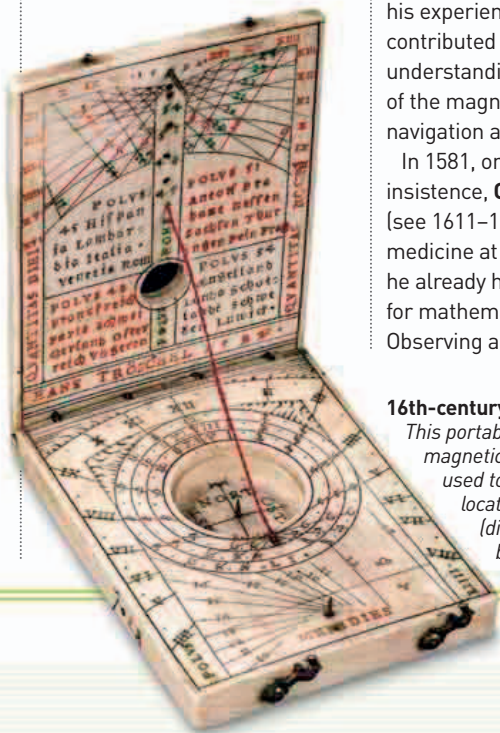
de Albacar. In 1581, Borough published his own treatise, dealing with the properties of magnetism and its effects on a compass needle. The treatise, reflecting his experience as a seaman, contributed considerably to the understanding and practical use of the magnetic compass in navigation and cartography.

In 1581, on his father's insistence, **Galileo Galilei** (see 1611–13) was studying medicine at Pisa, Italy. However, he already had a fascination for mathematics and physics. Observing a chandelier swaying



Galileo and the pendulum
Galileo first discovered the constancy of a pendulum swing by observing a swaying chandelier and timing its motion against his pulse.

in Pisa Cathedral, he noted that each swing took the same amount of time, regardless of how far it traveled. He then **experimented with pendulums** and found that the rate of swing was constant, no matter how wide the swing, and that two pendulums of the same length would swing in unison even if their sweeps were different. Galileo later published his observations on the constancy of pendulum swing.



16th-century magnetic sundial
This portable sundial has a magnetic compass that is used to align it in different locations. The gnomon (diagonal string) must be set north–south.

“...THE FORM AND COLOUR OF EXTERNAL OBJECTS... [AND] LIGHT ENTER THE EYE THROUGH... THE PUPIL AND ARE PROJECTED ON [THE OPTIC NERVE] BY THE LENS.”

Felix Platter, Swiss physician, 1583

BY 1582, THE JULIAN CALENDAR, which had been in use in Europe since Roman times, had become out of step with the times of the equinoxes by about ten days, so Pope Gregory XIII issued a decree introducing a new calendar. The Julian calendar had approximated the year, the time between successive spring equinoxes, as 365.25 days. This led to a discrepancy of about three days in 400 years. The reformed calendar, which came to be known as the **Gregorian calendar**, worked from a more accurate measurement of the time between the spring equinoxes. It was adopted first by the Catholic countries and gradually elsewhere.

Colonization of the New World gained pace towards the end of the 16th century. Writer **Richard Hakluyt** (1552–1616) helped to promote the English

Andrea Cesalpino
One of the foremost botanists of the 16th century, Cesalpino revolutionized the classification of plants.

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THE NUMBER OF PLANTS IN CESALPINO'S HERBARIUM

settlement of North America. In his 1582 publication *Divers Voyages Touching the Discoverie of America* and other later books, he pointed out the advantages of colonization, citing the possibility of establishing plantations for foods and tobacco.

Italian physician and botanist **Andrea Cesalpino** (1519–1603), who had been director of the botanical garden connected to the University of Pisa, developed the first scientific method of botanical classification in his *De Plantis Libri XVI* (The Book of Plants XVI), published in 1583. Cesalpino classified flowering plants according to their fruit, seeds, and roots, rather than by their medicinal properties.



1580 Sultan Murad III orders the Istanbul observatory to be destroyed

c.1580 Prospero Alpini describes two different sexes in plants

1581 Steven Borough publishes A Discourse of the Variation of the Compass

1581 Galileo Galilei observes constancy of pendulum swing in Pisa Cathedral

1582 Catholic countries, including Spain and Italy, adopt the Gregorian Calendar

1582 Richard Hakluyt publishes Divers Voyages Touching the Discoverie of America, promoting English settlement of America

1583 Andrea Cesalpino classifies plants according to their fruit, seeds, and roots in De Plantis Libri XVI

1583 Felix Platter suggests that the retina is stimulated by light